

Figure 3: QEMU runs the typewriter program

```
read_and_display:
    mov ah, 0; BIOS 10h function code for keyboard read
    int 0x16; BIOS interrupt for keyboard service
    ; Waits for a keypress.
    ; Now we have the ASCII code of the pressed key in AL.

    mov ah, 0xe; BIOS 10h function code for tty output
    ; Anything in AL will be displayed.
    int 0x10 ; BIOS interrupt for video service

    jmp read_and_display

padding: times (510 - ($ - $$)) db 0
BOOT_SIGN: db 0x55, 0xaa
```

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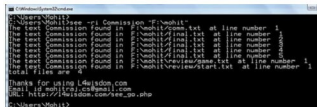


Figure 7: Option -r, which is case-insensitive

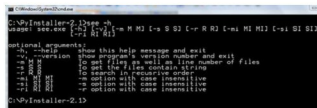


Figure 8: Help option

Use the option -si as shown in Figure 5. You can see that only file names are returned, and text searching is impervious to upper and lower case.

Use the option -mi as shown in Figure 6. Use the option -ri as shown in Figure 7.

In order to get help, use the option -h as shown in Figure 8.

The program offers you the power of regular expressions. Figure 9 shows the file 1.txt, which contains text. Let us use the regular expression, '+' operator.

The way ahead

All examples discussed in this article are based on the Intel x86 Real Mode, which has a limited memory of 1MB, and no protection walls between applications. Modern systems make use of the Protected Mode, which is a must to write a general-purpose OS. However, for backward compatibility and simplicity, modern CPUs still start in the Real Mode. That means the examples we discussed are still useful, avoiding the need for a complete reboot.

You can find many websites that can help you in all these processes, among which is the remarkable wiki.osdev.org. However, as the term *bootstrap loading* suggests, you are pulling yourself up with your own bootstraps, which means that original experimentation and a lot of patience are required. ed@nandakumar.com

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Figure 9: Regular expressions

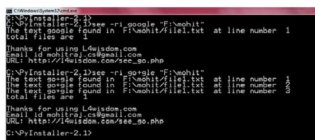


Figure 10: The power of regular expressions

See Figure 10, which shows the power of regular expressions. ed@nandakumar.com

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